

ActInf ModelStream 010.2 ~ RxEnvironments.jl with Wouter Nuijten

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all right hello it is May 16th 2024 we're in the RX and fur JL project with some guests from bias lab and participants and vzer will be sharing with us about the RX environments. JL package so thanks all for joining looking forward to this presentation and discussion go for it all right thank you Daniel uh so my name is V I am a second year PhD student in bias lab uh you may know Bas laab because we are the uh main authors of the RX infer package a package for reactive Beijing inference and last year when I was working with RX and fur um and doing well reactive beijan inference I found that there was a a lacking support for environments in the same reactive Paradigm uh and we were relying on reinforcement learning uh packages such as gymnasium or or reinforcement learning and I didn't really like it so I wrote RX environments. JL a package for reactive um environments which aims to uh complement RX and fur in the reactive um agent development um ecosystem so if we look at RX environments it's a very lightweight boiler plate package it contains a boiler plate code to make something that I call a reactive environment a reactive environment is a an environment that consists of entities where an entity can be an agent but it can also be a component of an environment or the entire environment itself and what we have to describe is how entities communicate with each other so every entity has its own Mark of blanket with which it can communicate with other entities in the environment so in that sense it's very um related to active inference because well we have a very specific notion of the interior of an entity and the exterior of the entity and it's Mark of blanket and I try to replicate that in ARX environments so um to give a short demo we can dive right into the code I have prepared um the mountain car example in the uh documentation of ARX environments but just to Showcase to you that this gives you um more um flexibility to work with so we have on the left hand side I have my code because I'm going to manipulate this environment uh in real time and on the right hand side I have the environment so I have my landscape I hope you guys can see my cursor by the way uh I have the landscape and in the middle I have one agent one car a DOT representing a

car um what's nice about RX environments is that I can send um actions to the environment so my landscape the environment is one entity the agent the car is a different entity and I can send a command that contains Engine Force on a scale from minus one to one so I can send uh minus one to The Entity and you'll see that the car starts driving up the hill it doesn't have a lot of Engine Force and I can live send different commands and you'll see that the car responds accordingly now what's nice about arcs environments is that as I said uh you'll get native uh multi-agent environments out of the box so if I just add a different agent Mountain car agent with some hyper parameters to the environment you'll see that there is now a second dot at the bottom of the valley a second car and similar to how this first car can be controlled we can control the second car too 01 was maybe uh this uh I should have maybe given this car a bit less engine power because it's clearly flying off the screen um but you get the gist um that we can um well control both cars simultaneously um and RX environments handles all this uh this communication natively reactively under the hood so my idea is or my next steps for this project are going to be to uh incorporate better into the ARX and fur ecosystem because I think that building agents uh with ARs environments in this way U that both the active inference Community as well as the arcs Ander Community can can can benefit from this so this is a very short overview of the things that ARX environments can do it can do a whole lot more um but I think that this is a very short demo that gives you the best idea of why it's nice that we're going to go reactive because you get stuff like this we can control in real time we can control multi-agent environments um and it's very um active inference e in that sense so on that note um I'll stop sharing my screen let's see and um yeah if you guys have any questions about this which I uh which I heard you have then I'll be very happy to take it awesome thanks very informative and hilarious Mountain car example yeah this this you you can play with this Engine Force of this mountain car and I probably should have prepared that it doesn't fly off the screen but you I mean it gets the point for us right it's yeah okay um I'll share screen and just type and so everyone else in the group just go for asking any of the questions we'd already identified or any new questions arising so go for it um interested Rel between I I I don't understand you very well maybe there's do you understand c as well okay relationship between you're you're very very faint uh yeah okay well with stuff I I actually had a question do not on this particular page that um Daniel's sharing right now I tried my hand at let's say plugging in the mountain car the existing Mountain car agent in the RX and thir documentation into your Mountain car environment I wasn't quite able to do that but it appears that you you have done that uh in that example just then and then maybe I was missing

something because in the in the documentation the RX environments documentation from mounting car I didn't see that there was a an agent that had you know instantiated the generative model uh much in the same way that the ARX infer documentation had and I thought oh okay this this is a wrapper or you know a nice API for me to specify an environment I I now have to go away and then specify my agent also but it seems is that what you did for that example just then or or did you just no no I just for this example just now I just used my um you know custom commands that I just uh sent myself um so this is something that I'm actively working on on uh combining the two um the mountain active inference Mountain car example on the ARX and fur website I believe is a bit um cheating in a sense because we give it all Environ Dynamics all functions of the environment Dynamics and then we uh do some uh some approximation to the posterior of what we have to do but in a sense it doesn't really discover that it has to do anything because it knows all the environment Dynamics so that's why I haven't really spent time on um incorporating this specific example because I feel that if we're going to demo this package plus an agent that it should be an agent that actually has capabilities of exploring the environment instead of knowing all the environment Dynamics and just while basically doing model predictive control right okay that makes it a bit clearer okay I I have a follow up maybe let somebody else um something just for now thank you very much so to give a very short answer to your question we can we can sit together if we uh on if want to Define an agent a generative model I know how to hoop the two up I just don't have a working example yet right okay I have I don't want to use this I don't want to use this agent from from the arxs and F documentation because indeed I feel that it's a bit cheating but it's my personal opinion right yeah yeah okay that that actually clears a lot of things off for me because I was kind of unclear like oh do I do I kind of plug that one in is that is that what I do yeah okay but that's not that functionality exactly CLE yeah we can um however there is a a subtlety there and that is something that I find very interesting in my research is if you open up this space or you open up your environment to be reactive then you want your environment to be able to um receive an action at any point in time so what we see in reinforcement learning is that you have a specific transition function the environment Waits until you send an action it will move the environment forward a pre-specified amount of time and it will emit to you an observation and a reward however if you make this entire thing reactive then well our agent would be able to emit at any point in time and also it would be able to get an observation at any point in time now that is a very interesting um setting to work with because now our agents should kind of be robust to getting an observation at any point in

time specifically what if uh our agent that lives on some embedded device on some drone whatsoever it has less compute to do to do inference however well at some point we're going to have to send an observ or an agent an action back to the environment we're going to have to do something so this reactivity opens up uh a research Avenue for well right now for me because I find find this very interesting on how reactive truly reactive agents would look like if you see what I mean yeah yeah I do that's that's exceptionally exciting stuff it's it's awesome um okay but but okay just just to wrap up my my yes question there is no at the moment you kind of manually you know yes you put in the force limit and you say okay I'm going to kind of manually you know play around with letters and I have I have working examples with like ban therat or very very easy um which I probably I don't know if I have them on my machine right now or if they live somewhere like it works but the the thing that I got to work is model predictive control and not active inference yet so that's why awesome all right thank you you're welcome um is my yeah can you hear me a bit better now yes okay but um I'm uh interested in the relationship between orx environments and orex in fur um for example to put it simply is it possible to use only orx environments or is it supposed to be always be used in collaboration with RX and fur us yes the relation is that uh I wrote ARX environment specifically um as a complimentary package to ARX and fur so if you just want to do basian inference you use ARX and fur but I think if you want to build agents that you're better off using the the two together so RX environments is um a package that uses the same reactive backend as RX and fur so it's made to be compatible with ARX and fur however if you write um agents just with an algorithm you can use ARX environments by itself it's not mandatory to use ARX and fur they are separate packages there's not a dependency on ARX Ander however since um they share the same reactive back endend so the same package for reactivity under the hood which is also from bias laab which is rocket um if you want to build agents I would advise to to use RX INF fur so there they're disjointed but they're designed with each other in mind or ARX environments is designed with ARX Ander in mind so um if we um use this high level of a of abstraction given to us by ARX Vons uh do we lose anything uh or can we at any time dive into the lower levels by means of orex INF fur itself so um I would say yes because the inference process that happens in ARX and fur is disjointed from what's going on in ARX environment similarly to well what happens in in in real life so you can you can see ARX environments as a package in which you can make um like a digital twin of the environment but then whatever happens in the inference process in your agents well that's still RX andur like all RX andur um diagnostic tools or debugging tools should be available there

um and a more practical question I know from based on the large papers um and uh Professor def as well they like to divide the um what's going on into these six methods the ACT future execute observe compute and slide so do you see that as kind of Superfluous in the orx orx environment setting or um is it is it fine to use that approach let me Google the definition for Superfluous uh ah um that is a I'm going to spread very carefully in answering that questions uh I don't think it is Superfluous um because in a sense that it's if you want to design your agent such that it goes through those steps um then you're more than welcome to uh my personal opinion is that I'm I will I'm going to try and see if I can make an active inference agent that does inference in one step instead of having to act observe uh plan slide uh and so forth however for AR environment so for your environment it doesn't really matter how you will get to your uh to your action as long as well or not right you can you can send an action at any point in time and if you don't send an action the world will just continue that and arcs environments will just continue running there's no real constraints on that so that the design is to be as less constraining on on users as possible while opening up this reactivity uh um the reactivity aspect and also for um if you start with the package you could make what I call discreet environments they're they're a bit feeding I would say because they're basically just the arc environment's equivalent of a of a classical reinforcement learning environment where you have a prespecify time step every time but I wanted to make that to make the transition easier because if you go full reactive right off the bat I would say it's a very ambitious thing to do so I don't think I don't think it's a Superfluous algorithm um but I I also don't think it's a um well our environment doesn't constrain you to use it so I would say nice now let's say uh one has a need to write one's own node in the factory um put ARX environments support that kind of thing or is it kind of an irrelevant question it's it's in that sense I would say irrelevant because uh the um a custom node is something that lives in ARX and fur um and ARX environments really uh is concerned with everything that goes on outside of your inference process um then another quick question while playing with the um I think it was a thermostat problem in oric environments I wanted to see the time behavior of some variables like observations States controls um free energy and things like that but it wasn't straightforward to me how to plot those um in the arment settings uh is there uh an example that you can point as to or a fragment of code that shows us how to do that um I can make it because I I know how I can access so free energy is uh something that lives inside your agent so it's part of the inference process it's the you know it's the uh cost function if you will of of of the inference process so for free energy you would have to look at your uh would have to look into ARX

and for however everything that goes on in our environments we can extract it and we can plot it so both uh environment States as well as emitted controls uh or observations uh we can extract all those things um I can show a code fragment that does this because it's well one line um however I also understand that that is not something that uh well you probably remember after this meeting so I'll I'll make sure that uh we can design a uh uh an example that showcases this but in uh in a nutshell we can uh create a keep actor so this is a well uh an entity that will keep all everything that it gets uh we'll keep it inside memory and then we subscribe to the observations of the agent so we say well subscribe to observations and with this Ops keep actor uh we do this and if I then call OBS values this will contain all kinds of well ARS environment observations which is basically a wrapper struct for uh let me call data on this entire array and you'll get your individual observation so we can basically tap anywhere in this process uh anywhere in this communication we can we can tap whatever is being emitted so if I run this again now it will contain 400 elements because well this environment is reactively running while we speak so the the the agent is getting more and more observations so you see this thing reactively growing every time I call uh or extract its uh its contents so there is definitely functionality in which you can sort of inspect this process um but yeah I I I have to stress that it's really distinct from the inference process so if you make an agent in ARX infer and you hook it up to ARX environments then you have to set up a similar procedure to extract free energy and stuff like that um from ARX and for um and because I'm not the author of ARX and for I I don't know how easy that is um but for AR environments I yeah this is one of the first things that I wanted to do see what my agent was sending see what my agent was observing so I made sure to build it yeah I'm not sure if what you just showed was uh recorded by us so if you could uh just make sure that that is included in your examples as well that would yes that that's my idea that we can uh or I I just wanted to Showcase uh for now such that you can so that you know that it exists but it wasn't meant to be well a big tutorial or anything I can just put it in the examples and uh uh yeah then then we can plot it that's that's not a problem okay thanks a lot you're welcome okay anyone else can go for a question or we have some other written ones or I can ask a question um I have a quick question I would just wonder once I don't know if you guys can hear me I hope so yes um fantastic um a lot of the examples that uh you have there and kind of the way I'm I'm probably naive in how I'm thinking about this um the ARX environment is modeling something in the real world um or at least that's the example that we're thinking of right now um is the is arc's environments capable of essentially modeling um more metaphysical spaces so if you think of

your question in a way that you have a Time series or a space that you're trying to occupy but it's not a physical space it's a more theoretical space uh you know dimensionality that you're trying to have an agent navigate um is that something that this that you see that this is you know usable for or is this something that we need to kind of gear back towards the agent for um package itself to kind of build the more metaphysical concept of the environment that you're putting together does that make sense so yeah if I understand your question correctly um yeah I my answer to that would be that there's no a physics simulator or whatever in arcs environments it's purely the boiler plate to make whatever you write act reactively uh and expose reactive streams that you can then use in arcs and so um for example one of our colleagues has made an environment that Returns the um the result of a stochastic differential equation so an agent could query with an action I want the realization at time T and the environment would respond with the actual value um and with this interplay of query in a stochastic differential equation and uh and getting a result you can cast solving a differential stochastic differential equation sort of a uh a Bayesian stat based model uh I'm not an expert on this topic I just know that one of our colleagues uh did this so if I understand your question correctly that would be the sort of environments you were after or you were you were questioning about um so I would say yes you can so there's no there's no physics simulator or anything in there because that's a whole different uh a whole different topic um my main thing was just to make whatever code you guys write to make it reactive fantastic thank you now that makes a lot of sense I just wanted to clarify that since no it's perfect not thinking so much in robotics or querying like physical environments and thinking more along the lines of data environment so that's why I was trying to understand how we could utilize this yes yeah you can yeah and uh you can push it further right you can make a uh a combination of the two you can make a uh an environment that flies a drone uh your agent has to control the Drone but it can also query the environment for hey what is going to be my position in 5 seconds so kind of this this well abstract action that's basically querying the environment well since we have this uh this mechanism that we can depending on what action you send correspond or match this with an observation now you can do stuff like that there's nobody stopping you doing it whether or not it's useful for your agent well that's up to you I I was just yeah I'm I'm trying to make your environments as expressive as can or your the framework as expressive as I can and yeah you you can you can go wild with it right you can go crazy that's perfect because the agent that I'm kind of that I'm wanting to put together would need to actually walk through time in a certain way um and optimize its activities in a Time series um yeah and I was I was

already expecting you that's why I brought up this this uh that's very cool I be very excited to see uh well whether or not you uh you can build your environment and ARS environment would be yeah very cool Absolutely I'll keep you updated then nice thanks the observable and actor roles in orx environments can any entity take on either of these roles or are there certain constraints for example let's say the agent must always have the actor role or something no they're completely unconstrained um there is no real distinction under the hood between uh entities or between something that represents an environment as something that represents an agent uh they are both entities in arcs environments um the only thing is that you can make uh something which I call an active or a passive entity by the way all this terminology it's all up for discussion I've thought about this and most of the terms that I come up with they they don't really fit what I what I'm trying to say um but an active entity is something that whenever it gets an observation or whenever it receives an action it emits or it at least considers emitting to uh subscribers for example the mountain car uh environment itself would be an active entity whenever the the agent sends an action to move the car it will correspond or it will match this uh this this action with well moving the car um whereas the agent whenever it gets an observation from the environment well it doesn't really have to do anything we have to in the later stage attach an inance process but that is something that's not ARX environments that is ARX and first so we we make this entity um passive in a sense that it it's it's an observer uh that can also send actions but those are not triggered by an actual algorithm that we write ourselves uh in arxs environments but it's a result of some inference process it's the triggers are external to the the reactive ecosystem that we've built in ARX environments itself so that's the only distinction that I make but you can yeah something that represents an environment or something that represents an an agent they're they're under the hood completely the same things they are just uh entities that that that can communicate with with with external entities do you think you could um in addition add a a discrete control space example for us as well because Mountain car and the evasion thermostat are both um continuous examples even if it's just a simple uh discrete control space or categorical in particular categorical control space maybe a simple maze example or a grid example you know like the windy grid that's often used in reinforcement learning even if it's just a simple windy grid windy grid yeah okay I'll I'll Google that just uh even if it's just a simple categorical control space because that's what our problems currently uh center around not so much continuous control spaces but uh categorical control spaces okay that's uh yeah I can do that okay I windy G world yeah okay I I can do that be excellent I ask sort of a very quick broad question about the whole ontology

my when I first learned about this my own Na and and just sort of looking through the docs I thought oh okay this is this is a rapper around RX and thir for standardizing the creation of an Environ enironment that you will then use for for whatever purposes but it sounds to me it's it's not at all that it's it's really this is this is not even really a way of talking about an environment per se this is no it's more of a it's a communication protocol for active inference agents I would say yeah yeah it's like well you this this is a way to sort of standardize the the dietic interaction between you know two abstract entities U and that could be between environment and an agent you know or it could be something else entirely okay that's much much broader than what I was conceptualizing ex so I I when I when I started this indeed I I started as okay I want to standardize environment Creation in in our ecosystem in our fer and I had discussion with Magnus kudal who's also uh in in this Discord I saw and well we had some very interesting talks and he said well actually an environment and and an agent they're not so different right they just they just hand or they just receive data from each other and and handle them accordingly so yeah there is a certain uh symmetry there and then I thought well if there's a symmetry there then I can make an overarching uh programming concept and just kind of implement it that way and that's what I did and then it turns from just a way in which you can make an agent and an environment communicate to a in which you can communicate this whole network of uh of subscriptions and Communications um because there's yeah there's no constraints anymore so for example we run an environment in in uh in our lab with MCO actually that um has well a non-trivial uh entity interaction where we have a hearing aid that uh gets data from both the external world because it can well it has a microphone and it should register the external world but it can also get data from the user of the hearing aid saying hey I'm happy with the hearing aid settings yes or no so now we have a three entity environment we have the user we have the hearing aid and we have the uh the external world that just emits noise then we have a fourth entity which is our agent that should learn the preferences of the user that should get data from the hearing aid and should uh tweak the parameters inside of the hearing a so now we have four entities that are communicating in this non-trivial way and well we built this in arcs environments and it all holds up like it all just yeah it runs just because whenever we get data the hearing aid knows ah I get new data I get new noise I have to transform this noise and send it to the user and I have to do a f8 transform and send it to the agent for analysis uh and just because it does this reactively it's all tractable and all works nicely so yeah uh it it it turned out to be a bit of a uh a bigger well scope than I would have imagined but I'm yeah I I'm I'm not making any claims on on how large it actually

is that's uh yeah cool this this is also very cool I'm very very thankful that you're you're working on this because I I I had the thought like oh what what what then is you know sort of predicated of the environment it's kind of like what we have the environment and the Asian they're kind of both being equally treated you it's a very interesting question right so so what the what is the actual actual difference um some people say well an agent has some sort of inferential process uh on how it comes up with an action but in a sense well we are all all just with our eyes and ears and muscles we have an interface to to the environment um but yeah in in in that sense we we are just all Mark of blankets with internal states that that that navigate this world uh and this well communication that we have with our outside world I yeah I try to replicate fastic thanks anyone else want to ask any questions yeah Kristoff or yeah yeah I have a question um I I didn't quite get what you um what you said about the difference between the typical reinforcement learning setting and the RX environments Could you um could you could you explain that again please that was very interesting sure sure if if you want or if you make an environment in JY in gymnasium we can we can look at the um at their documentation right so let's see gymnasium package so I go to gymnasium and I want to make my own custom environment I saw it here make your own custom environment i' I've visited this page before so we can make our environment blah blah blah and then you have this function step the step method usually contains most of the logic for your environment it accepts an action computes the state of the environment after applying that action and returns a five Tuple observation reward terminated truncated info so we write a function step that takes only an action so this is the main design Paradigm of JY and most reinforcement learning packages actually where I emit an action this action is incorporated into the world um but the time that uh the environment is uh simulated forward is constant so there's no notion of different time intervals whereas if I'm I'm a very big uh football effci out for the Americans that's soccer um but it's actually football but if I kick a ball I kick the ball the ball starts rolling well whether or not I do something the ball will just continue rolling like this this environment around me it doesn't wait for me to emit additional actions or I'm not consciously saying okay for the next second I do nothing and then the ball rolls okay for the next second I do nothing again and the ball rolls further no there is a in the real world there is a difference right we do something and then the world just continues so we can do something at any point in time and the world around us will just well the world keeps on turn um and that's what I was trying to do with with ARX environments is to um alleviate ourselves of this um communication protocol because this specifies I get one observation I can do one action per observation and

that's it and in between uh different observations I have no power to do anything whatsoever also if I have an environment that instead of modeling one second forward I now do half a second well everything breaks down because now I have to emit actions at twice a speed or my observations don't correspond anymore with my internal model or there's all kinds of stuff that if you do it like this it turns kind of awkward and that's what I was trying to alleviate uh with with ARX environments is that I'm trying to make this communication between agent and environment as free as possible so if you have a hearing aid environment yeah that you can yeah that's that that's it if I'm not yeah I'm rambling sorry good interrupt that's a good that's a good explanation and U I think you also mentioned something about um some difficulty there because uh the the question then becomes like if you have multiple agents in with the environment then what exactly is the state of the world right um what exactly is the state of the world is not uh because you can just uh sort of you can have like a god entity that takes care of of of this stuff that that that kind of resolves all conflicts in that sense um the main question for me would be well we just took away your one moment to conduct an action in reinforcement learning you have one moment you can do an action and you'll get one observation we just took away this one moment and gave you the opportunity to act at any point in time well when are you going to do it right so you know the reason I'm asking about this is that it immediately got me thinking about um stepen wolfram's work on Observer Theory so he's trying to formalize uh the notion of an observer right and um there is this um whole field that he basically invented uh for the physics project related to uh which calls multi multi computation I'm not really deep into um that yet but it seems some it seems somehow relevant you know as an agents interacting with the environment we effectively sample it precisely because it keeps rolling as we um do the internal computation required to um update our um our generative model in order to to act so I I think you stumbled onto something quite quite profound there um and the question the question is about this this God God mode view right like that's a that's an old question in physics whether there exists the absolute reference frame or whether everything is relative um and yeah just won't say that you know foot for fault there's there's some there's some interesting problems there maybe we can we can talk about it some sometime I would love to hear your thoughts about this yeah sure like I'm I'm very interested in this stuff so as a as a software developer like I would advise you in your environments if you're not dealing with you know existential stuff that just make a god state that takes care of this but uh you make a very good point right so if we really want to make an accurate representation of of the world do we need this God state or because we now allow our

environments to to alleviate this right we can have multiple entities that together compose the environment and that there is some internal communication between them so um it does it does open up this like it doesn't slam it wide open but at least you know there's a in Dutch you would say open key yeah so so I I think there is a there's a that that's that's a potential um incursion into effectively resolving some of the most interesting problems in the fundamental theories of physics that understanding precisely which one is the case whether is this um absolute reference frame or whether you can just simply explain everything through these relative reference frames and agent sampling the environments and if if you can effectively recover something that looks like physics from um this interaction which is what wall from is essentially working on that would be that would be a complete computation of fundamental Theory of physics and it's a bit big undertaking but it's worth uh worth considering not just merely from you know um software engineering perspective but from from that angle if if you're interested in that kind of stuff I mean yeah it it's very interesting it's something that I you you know it's a very big deviation from what my supervisor wants me to work on so it's probably not something that I'm going to undertake but yeah if if anyone wants to pick it up right who am I to stop them thanks it's great work cool yeah anyone else want to ask any questions how would you deal with the setting of multiple ant nestmates moving around and modifying the pheromones on the ground so in these kinds of simulations it's common enough to have the pherone phermone state on the ground and then Nest mates that are reading and writing how do you deal with the spatial proximity and the collisions and interactions so that is something very very very difficult I'm working on a project right now where once again we're modeling football uh which is why I brought up this example just now um and like resolving collisions is something extremely difficult it's something that I don't have a standardized uh advised way of doing an arcs environments just because game studios uh they they spend God knows how many developer hours on resolving collisions efficiently so that's not something that I think well we we can figure out in an afternoon right um it's it's it's something that would be in the internal dynamics of your environment but not something that is part of your communication protocol per se just influences the the the type of observations that your agent would get um so it's not something that I can give that I feel qualified to give advice on but it is like making this environment with ants running around the floor that is something that's just a multi-agent environment that's fully covered the problem is these these uh the collisions that is a difficult thing it's not uh difficult in an arcs environment uh perspective it's difficult in a computational geometry perspective that you want to resolve collisions that's yeah like in in

trip a games you can still clip into walls somewhere that's that's a great and funny response yeah I would just highlight the the real structural difference when you pointed to the step function in the gymnasium that yes is why in so often for these control problems there's like a null action or like a weight action so you kind of get this um bodal action Vector policy Vector where there's like a do nothing option and then all the options because out of requirement something must be emitted at every click of the simulation and all of a sudden it's like we mve from walking on a staircase to like ice skating or flying with RX and fur because it has totally opened up a dimension of the control question when to act not just which with according to a metronome and that's that's an entirely different approach that um the communication protocols enable and then those communication protocols can have all these other functionalities like they can have a subontology or they can have compliance features as well so it's super powerful and we're very excited to learn it that's yeah that's the main point uh the yeah it's a two sides of the same coin like it it opens up this communication protocol but then yeah you have to question when when it's something something that I'm going to spend time on in my next project like can we make an iterative procedure maybe with iterations and ARs and third we can send intermediate posteriors and we have we always have like a an action that we Supply I I don't know yet I have no idea or I have ideas but I don't have anything working yet I think to me the underlying principle of the RX in first Suite versus traditional yes this um um instead of having synchronized Communications we now have asynchronized communications between everything between everything yes I'm I'm glad I got that point across that's that's the point I wanted to make great well um if anyone else has a question otherwise we will continue learning and watching the GitHub updates and just building out examples CU in the coming months we know a lot of fun things will happen all right yeah um you guys can always reach out to me on I don't know GitHub Twitter I don't know how often I'll check Discord I'll try you know I'll try um but you can always reach out to me on uh on LinkedIn Twitter whatever that's that's all fine excellent I I do have a a final question if that's all right um sure just just with respect to you know future directions and that kind of thing what's kind of biggest problem set of problems what are you kind of looking to do I know that's a massive thing but is there anything relatively summited in that direction that that we could have speak to for just a second so I'm very interested in this when to act uh problem so this to me opens up the door to a um an agent that um because we okay so we can now act at any point in time or at any point in time um yeah we can send something to the environment a logical consequence for me is that at any point in time we should have an action ready to send

so when the next observation comes in we don't know when that's going to be we should be able to act so we have a certain computational budget between one observation and the next we don't know how much it's going to be but we do have a budget to spend so I want to design indeed an interactive procedure that comes up with an action and then refines it and refines it and refines it until well the next observation comes in so we we have to send whatever we had and we get the the next observation this gives you if this all works right this gives you a very nice robust agent that if you give it a lot of computing power a lot of computational budget then in between observations well it's going to be able to come up with a pretty good um pretty good action and will solve the task accordingly like ideally however if you limit this uh uh this computational budget you put it on like a Raspberry Pi or on an edge device it will not have as much iterations to refine its actions so it will still solve the problem hopefully but it will be less efficient doing so it will be slightly Dum so this is a very to me is a very interesting problem to think about because then you have a sort of a procedure for control that is robust to loss of computation or whatever and it's still adaptive and reactive and and all those nice things we get from from Asian inference um so yeah that's something that I that I would want to work on if I you know at 48 hours in a day it's very interesting so it sounds to me like a number of iterations you just refer to as um the iteration field in the call of the infer function yes in infer yes if I can do something like that so I run an indefinite amount of iterations and I just stop the inference procedure whenever I have to do something and then I extract the latest posterior for the action and I I send that to the environment yeah that's something well that would be the first thing I would investigate yeah I I think it has the potential to make the behavior in general quite robust if you if you consider that approach it yeah it would be rob it would always have something to do do like it would always do something whether or not sort of the the the free energy would converge uh or yeah as as a measure of whether or not this action is good enough uh that is to be investigated but it would allow us to run the same models on edge devices as well and get slightly worse performance but still yeah yeah it would still do the job we and then you can investigate you know how many iterations do I actually need before I can solve the problem uh it's of course problem dependent but yeah it's a very yeah it's just something that that really interest me I really like this this way of thinking um and it's in part why I why I wrote this package because I I don't like this this this synchronized communication protocol from Jim I just I just don't like it yeah that's why I wrote wrote ARS environments I think the only alternative is to not be able to respond whenever it gets an ation other words to postpone your response it NES back into the

um the the synchronicity approach again so it breaks away from the you know the reactivity of the but then you you run into the robustness issue I would say if you have a self-driving car and it it's driving into a wall because it's thinks it's a green light because those things are not that that smart nowadays uh and ah it figures out it's a wall but it takes some time to adjust or it takes some time to come up with the idea okay now I have fully converged on the action that I have to steer away well by that time we might have already crashed into the wall um so this this queuing up of uh uh of observations and waiting until uh my computation is fully converged I think it takes away the robustness like if you if if you don't have enough computational budget to spend well yeah you you might do the wrong thing it's the same as when you and I are in a car right if we're in a very um a very busy situation so well I'm in the Netherlands so we have cyclist right so if we have cyclists we have other cars and there's a an ambulance coming what it's very busy you might feel overwhelmed but you still kind of know the basic things to do to to to steer your car it's basically the same where you have limited computational budget there's all this stuff around you so to come up with a really thought through response or action uh it's too much to ask because you don't have the computational budget but your brain kind of prioritizes and says well if we stay on the road and we let the ambulance pass and well we're fine for now we're this is the best we can do um so this feeling of being overwhelmed by what what goes on on around you inspired this idea of of robustness and uh uh of interruptibility basically yeah but then again how how how I'm going to do it well no idea yet or run run our inner iterations and see if I can uh can send intermediate posterior results or something or but it's just something yeah I I find this very very very interesting think it's a good thing that free energy in general decays uh predictably in a sense so I mean just fully using your budget your computational budget for that time that you're allow before the next observation arrives I think it's it's fairly safe to just uh yes use the current value of the posterior that's the best you can do the circumstances yes that that that would be the argument indeed hopefully hopefully this this this opens up the road to having actual you know robots driving around instead because well there's no self-driving cars where I live there's no robots or drones flying around and I think one of the reasons is because they're not robust or interruptible fly agreed yeah all right thank you all we'll look forward to staying updated asynchronously even asynchronously reactively see y all soon bye thanks